Unit 2 Review

Name _____

One study, every tool. A class flew paper airplanes. Here are 5 flights, in feet.

14, 18, 22, 22, 24

2.1 Find the SD.
The SD is about how far the values usually are from the mean.

$\bar{x} = 100 \div 5 = 20$ feet

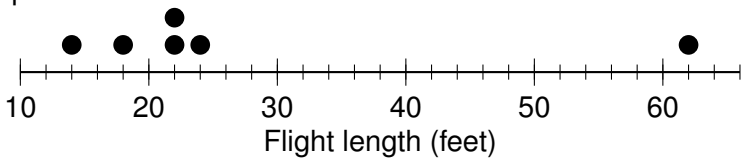
x	14	18	22	22	24
$x - \bar{x}$	-6	-2	2	2	4
$(x - \bar{x})^2$	36	4	4	4	16

$\Sigma(x - \bar{x})^2 = 64$ $\quad s = \sqrt{\dfrac{\Sigma(x - \bar{x})^2}{n - 1}} = \sqrt{\dfrac{64}{4}} = \sqrt{16} = 4$ feet

The flights are usually about 4 feet from the mean.

2.2 Choose the measures.
Symmetric, no outliers: mean and SD.
Skewed or outliers: median and IQR.

A sixth plane flew 62 feet.



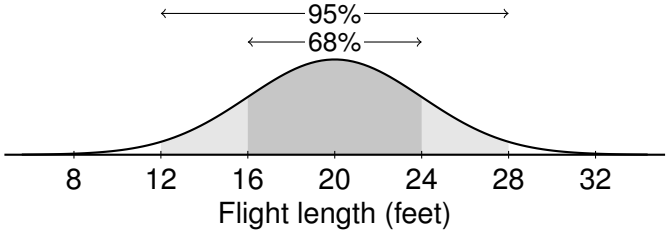
Mean = $162 \div 6 = 27$ feet. SD ≈ 17.5 feet.
Median = 22 feet. Q1 = 18, Q3 = 24, IQR = $24 - 18 = 6$ feet.
Upper fence = $24 + 1.5 \times 6 = 33$, so 62 is an outlier. It pulled the mean from 20 up to 27. Use the **median and the IQR**.

2.3 Where does one value stand?
Percentile: the percent of the data below the value.
z-score: how many SDs the value is above or below the mean.

4 of the 5 flights are shorter than 24 feet. $4 \div 5 = 0.80$. So 24 feet is at the **80th percentile**.
All the school's flights: mean $\mu = 20$ feet, SD $\sigma = 4$ feet.
A 26-foot flight: $z = \dfrac{x - \mu}{\sigma} = \dfrac{26 - 20}{4} = 1.5$. It is 1.5 SDs above the mean.

2.4 Use the empirical rule.
In a normal distribution, about 68% is within 1 SD of the mean, 95% within 2 SDs, and 99.7% within 3 SDs.

The school's flights have a normal distribution.



68% of the flights are from 16 to 24 feet. 95% are from 12 to 28 feet.

2.5 Use the z-table.
The table gives the area below z.

Shorter than 25 feet: $z = \dfrac{25 - 20}{4} = 1.25$. Table: .8944. **89.44%**
Longer than 25 feet: $1 - .8944 = .1056$. **10.56%**

2.6 Work in reverse.
Find the area inside the table. Read z. Then $x = \mu + z\sigma$.

The longest 10% of flights win a prize. Area to the left: $1 - .10 = .90$. Closest area .8997, at $z = 1.28$. $x = 20 + 1.28 \times 4 = 25.12$. A flight must go about **25.1 feet**.

Above or top? Subtract from 1. The z-table always gives the area below z (to the left).

Work with a partner. Compare answers when you finish each problem.

2.1 - Standard Deviation

1. **I find the standard deviation of a small data set.**

   A bike shop sold this many bikes each day for 5 days.

   4, 11, 7, 11, 7

   (a) Find the mean, $\bar{x}$. Then complete the table.

   $\bar{x}$ = _____

x	$x - \bar{x}$	$(x - \bar{x})^2$
4		
11		
7		
11		
7		
	$\Sigma(x - \bar{x})^2$	

   (b) Find the standard deviation, s. Round to 2 decimal places if you need to.

   Four friends scored these points on a 10-point quiz.

   9, 5, 8, 6

   (c) Find the mean, $\bar{x}$. Then complete the table.

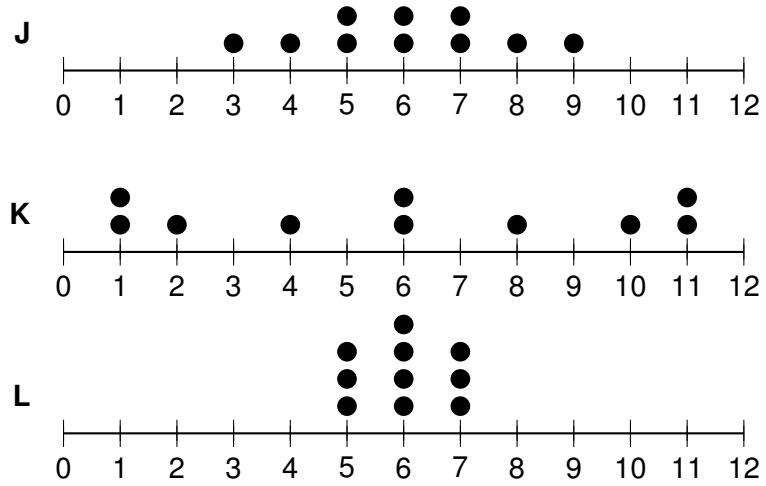
   $\bar{x}$ = _____

x	$x - \bar{x}$	$(x - \bar{x})^2$
9		
5		
8		
6		
	$\Sigma(x - \bar{x})^2$	

   (d) Find the standard deviation, s. Round to 2 decimal places.

2. **I compare the spread of data sets by looking at graphs, and I explain what the standard deviation tells about data.**

Each dot plot has 10 values and a mean of 6.



(a) Put the dot plots in order from the **smallest** SD to the **largest** SD.

(b) Explain your order.

(c) The SD of plot K is about 3.94. What does that number tell you?

(d) A data set has an SD of 0. What do you know about its values?

(e) Set M is 40, 45, 50, 55, 60. Set N is 48, 49, 50, 51, 52. Both have a mean of 50. Without calculating, which set has the larger SD? Explain.

(f) Can a standard deviation be negative? Explain.

2.2 - Choosing Measures and Comparing Distributions

3. **I choose mean and SD or median and IQR based on the shape of the data.**

For each data set, circle the measures you should use. Then explain.

(a) The heights of 500 corn plants in a field. The histogram is symmetric, with no outliers.
mean and SD median and IQR

(b) The number of followers 40 students have on a social media app. The histogram is skewed right.
mean and SD median and IQR

(c) Scores on a very easy test. The histogram is skewed left.
mean and SD median and IQR

(d) The times 200 students took to run a mile. The histogram is symmetric, with no outliers.
mean and SD median and IQR

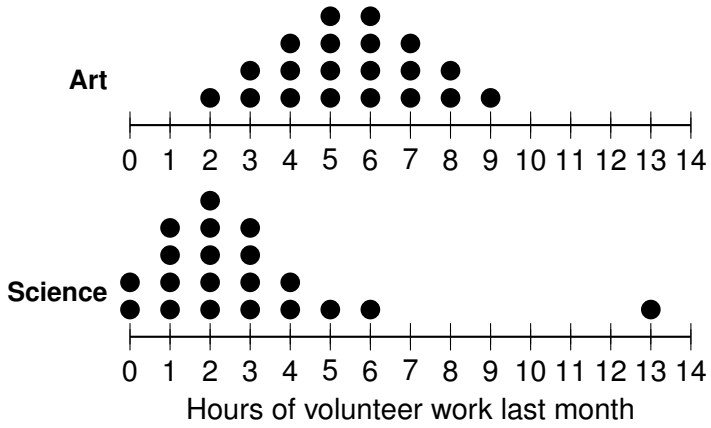
(e) Minutes 7 students spent on homework: 30, 35, 40, 40, 45, 50, 120
mean and SD median and IQR

(f) High temperatures for 9 days, in °F: 70, 72, 73, 74, 75, 76, 77, 78, 80
mean and SD median and IQR

(g) For the homework data in (e), find the mean and the median. Which one better describes a typical student? Explain.

4. **I compare two distributions using shape, center, spread, and outliers.**

These dot plots show how many hours students in two clubs volunteered last month. Each club has 20 students.



	Q1	Median	Q3	IQR
Art	4	5.5	7	3
Science	1	2	3.5	2.5

(a) Describe the shape of each club's data.

(b) Compare the centers. Use numbers and the context.

(c) Compare the spreads. Use numbers.

(d) Does either club have an outlier? Explain.

(e) Should you compare these clubs with the mean and SD or with the median and IQR? Why?

Two pizza places recorded their delivery times. Both histograms are symmetric.

	Mean	SD
Place A	30 minutes	3 minutes
Place B	26 minutes	8 minutes

(f) Which place is faster on average?

(g) Which place has delivery times that are more spread out?

(h) Write one sentence that compares the two places. Use both the center and the spread.

2.3 - Percentiles and Z-Scores

5. **I find the percentile of a value.**

20 students said how many minutes they practiced a musical instrument yesterday. Here are the times, in order.

10, 12, 15, 18, 20, 24, 25, 28, 30, 32, 35, 36, 40, 42, 45, 48, 50, 55, 60, 90

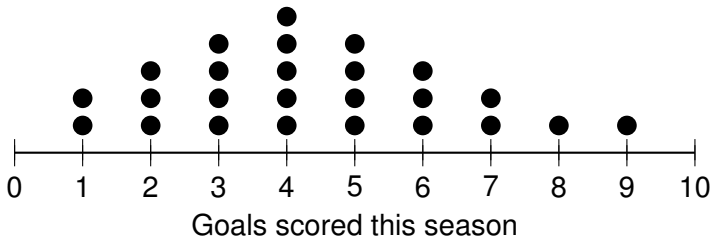
(a) What percentile is 40 minutes?

(b) What percentile is 15 minutes?

(c) Which time is at the 25th percentile?

(d) Which time is at the 90th percentile?

This dot plot shows how many goals 25 hockey players scored this season.



(e) What percentile is 6 goals?

(f) What percentile is 3 goals?

(g) Nadia says, “My test score is at the 90th percentile, so I got 90% of the questions right.” Explain her mistake.

6. **I find and explain a z-score, and I compare values from different data sets with z-scores.**

Scores on a spelling test have a mean $\mu = 50$ and an SD $\sigma = 6$. Find the z-score of each score. Then say what it means.

(a) 62

(b) 47

(c) 50

(d) 41

Omar and Lena compete in different events.

Athlete	Event	Jump	Mean μ	SD σ
Omar	long jump	5.6 m	5.0 m	0.3 m
Lena	high jump	1.55 m	1.40 m	0.10 m

(e) Find each athlete's z-score.

(f) Lena jumped 1.55 m and Omar jumped 5.6 m. Who did better compared to the other athletes in the same event? Explain.

Go back to the spelling test ($\mu = 50$, $\sigma = 6$). Use $x = \mu + z\sigma$.

(g) What score has a z-score of 1.5?

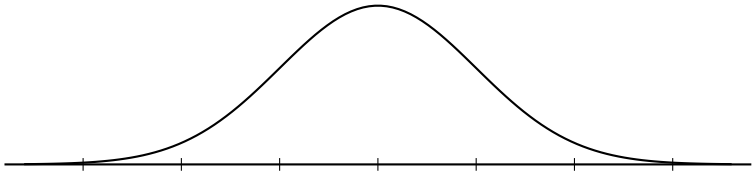
(h) What score has a z-score of -2?

2.4 - Normal Distributions and the Empirical Rule

7. **I use the empirical rule (68-95-99.7) to find percents, and I find the percentile of a value 1, 2, or 3 SDs from the mean.**

   The heights of sunflowers in a field have a normal distribution. The mean is 180 cm and the SD is 12 cm.

   (a) Label the number line with the heights.



Height of a sunflower (cm)

   (b) About what percent of the sunflowers are from 168 to 192 cm tall?

   (c) About what percent are from 156 to 204 cm tall?

   (d) About what percent are taller than 192 cm?

   (e) About what percent are shorter than 156 cm?

   (f) About what percent are from 180 to 204 cm tall?

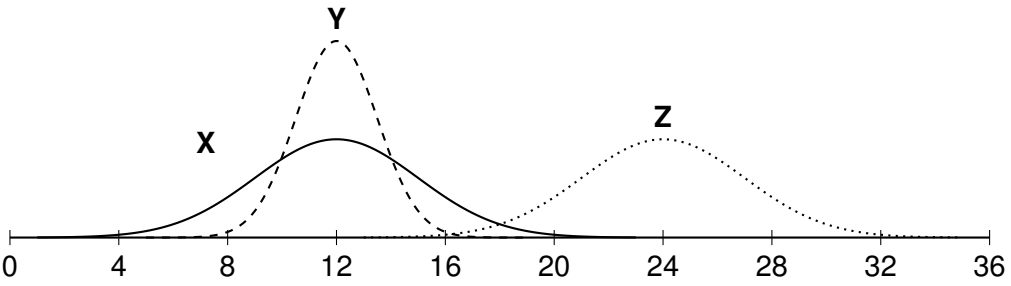
   (g) About what percent are from 192 to 216 cm tall?

   (h) What percentile is a height of 204 cm?

   (i) What percentile is a height of 168 cm?

8. **I describe a normal distribution, and I compare normal curves with different means and SDs.**

The three curves are normal distributions.



(a) Which two curves have the same mean?

(b) Which two curves have the same SD?

(c) Which curve has the smallest SD? Explain.

(d) How are X and Z the same? How are they different?

Would each data set have a normal distribution? Explain.

(e) The heights of 1,000 adult women.

(f) The prices of every car sold in a city.

(g) Scores on a very hard test.

(h) Which statement is true for every normal distribution? Circle the letter.

A About 95% of the data is within 1 SD of the mean.

B A larger SD makes the curve taller and narrower.

C The curve is symmetric about the mean.

D The median is greater than the mean.

2.5 - Normal Calculations with a Z-Table

Use the z-table.

9. **I read a z-table.**

Find the area **below** each z-score.

(a) $z = 0.36$

(b) $z = -2.15$

(c) $z = 1.82$

(d) $z = -0.07$

Find the area **above** each z-score.

(e) $z = 0.95$

(f) $z = -1.38$

Find the area **between** the two z-scores.

(g) $z = -2$ and $z = 2$

(h) $z = 0.25$ and $z = 1.75$

10. **I find the percent of data below, above, or between values in a normal distribution.**

The lengths of movies at a theater have a normal distribution with a mean of 110 minutes and an SD of 12 minutes. What percent of the movies are:

(a) shorter than 125 minutes?

(b) longer than 128 minutes?

(c) from 100 to 120 minutes long?

Bags of trail mix have weights with a normal distribution. The mean is 250 grams and the SD is 4 grams. What percent of the bags weigh:

(d) less than 245 grams?

(e) more than 251 grams?

(f) from 247 to 256 grams?

2.6 - Normal Calculations in Reverse

Use the z-table.

11. **I find the z-score for a percentile with the z-table, and I find the value at a percentile in a normal distribution.**

    Find the z-score at each percentile.

    (a) 72nd percentile

    (b) 15th percentile

    (c) 99th percentile

    (d) 42nd percentile

    The times to finish one level of a puzzle game have a normal distribution. The mean is 90 seconds and the SD is 15 seconds.

    (e) What time is at the 72nd percentile?

    (f) What time is at the 15th percentile?

12. **I find cutoffs for "top" and "bottom" percents.**

    Typing speeds in a class have a normal distribution. The mean is 40 words per minute and the SD is 6 words per minute.

    (a) The top 2% get a certificate. How fast must a student type to get one?

    (b) Students in the bottom 25% get extra practice. Below what speed does a student get extra practice?

    The weights of pumpkins at a farm have a normal distribution. The mean is 12 pounds and the SD is 2 pounds.

    (c) The heaviest 10% are sold as "giant" pumpkins. How heavy must a pumpkin be to be giant?

    (d) The lightest 20% are used to make pies. Below what weight is a pumpkin used for pie?

Test-Style Questions

A test asks the same questions with more words. Read slowly. Circle the best answer.

13. Each data set below has a mean of 50. Which data set has the greatest standard deviation?

    A 48, 49, 50, 51, 52

    B 30, 40, 50, 60, 70

    C 45, 50, 50, 50, 55

    D 50, 50, 50, 50, 50

14. A restaurant recorded the number of minutes that each of 50 customers waited for a table. The distribution of the wait times is strongly skewed to the right and has several outliers. Which measures of center and spread are most appropriate to describe the wait times?

    A the median and the interquartile range

    B the mean and the standard deviation

    C the mean and the interquartile range

    D the median and the standard deviation

15. On a science test, the class mean was 72 points with a standard deviation of 6 points. On a history test, the class mean was 80 points with a standard deviation of 5 points. Mia scored 84 points on the science test and 88 points on the history test. Which statement is true?

    A Mia did better on the history test compared to her class, because 88 is greater than 84.

    B Mia did equally well on both tests compared to her class, because both scores are above the class mean.

    C Mia's history score was 1.6 points above the class mean.

    D Mia did better on the science test compared to her class, because her z-score was greater on that test.

16. The lifetimes of a type of light bulb are approximately normally distributed with a mean of 900 hours and a standard deviation of 50 hours. According to the empirical rule, approximately what percent of the light bulbs last between 800 hours and 950 hours?

    A 47.5%
    B 68%
    C 81.5%
    D 95%

17. The scores on a math contest are approximately normally distributed with a mean of 60 points and a standard deviation of 8 points. Use the z-table. What proportion of the scores are greater than 74 points?

    A .0401
    B .4599
    C .5401
    D .9599

18. The weights of watermelons at a farm are approximately normally distributed with a mean of 20 pounds and a standard deviation of 2.5 pounds. The heaviest 3% of the watermelons win a blue ribbon. Use the z-table. Which weight is closest to the lightest watermelon that wins a blue ribbon?

    A 15.3 pounds
    B 22.5 pounds
    C 24.7 pounds
    D 25.0 pounds